

Judge's Commentary: The Outstanding Faculty Salaries Papers

Donald E. Miller

Dept. of Mathematics

Saint Mary's College

Notre Dame, IN 46556

dmiller@saintmarys.edu

The substance of this problem was recognizable by the judges from academic institutions, causing many of them to comment on how similar it is to the situation at their own institutions. The basic problem of designing a new pay system satisfying the specified criteria was simple enough that most teams recognized it as a curve-fitting problem. They also, with varying degrees of success, were able to find a model that satisfied most of the criteria.

Implementing the model, however, with limited resources, on salaries of current faculty at Aluacha Balaclava turned out to be deceptively more difficult. Most teams recognized that the salary curve would need to have a negative second derivative in order for salary increments to decrease with experience. Models included variations of logistic, power, exponential, root and polynomial functions, with the exponential function appearing most often. In their Outstanding paper, the team from University Alaska Fairbanks indicated that they had experimented with six different functions before settling on the logistic model because it met all the criteria.

Some teams used an appropriate model for the first twenty-five years of experience, then froze everyone's salary, thus violating the principle that all faculty should get a raise any year that money is available. Most teams used the same model for all four ranks and entire tenure of the faculty member. However, others recognized the instructor rank as different from the other ranks and developed a separate model for it. Still others developed a separate model for the full professor rank, arguing that it was different because a full professor has no chance for promotion. The Outstanding paper from Southeast Missouri State University even developed a separate model for each rank.

Modelers can frequently gain creditability by demonstrating that they understand the problem in its context. With our salary data, this could have been accomplished by recognizing some of the salaries as outliers that would need to be dealt with individually. The datum point representing a professor with two years of experience and a salary of \$85,500 is clearly an

outlier and will be virtually impossible to bring into line with any reasonable model. This person may be some sort of a “superstar” who is not expected to fit into the model salary structure. One of the judges suggested that this salary belonged to the football coach. Whatever the case, the modeler should be willing to point out or question such unusual situations, but few of the teams did. One exception was the Outstanding paper from the North Carolina School of Science and Mathematics. This team showed that, “...with the exception of a few grossly overpaid faculty, the problem of unfairness would be solved” in five years.

Better papers were distinguished by a complete and mature treatment of the assumptions as well as a precise plan of implementation. The team from University of Alaska at Fairbanks even offered two plans, showed graphically how they differed, and then recommended one over the other. All the Outstanding papers also did some sensitivity analyses and included cost of living in their implementation plan. Another feather of these papers was a well-written executive summary.

About the Author

Donald Miller is Associate Professor and of Chair of Mathematics at Saint Mary’s College. He has served as an associate judge of the MCM for three years and prior to that mentored two Meritorious teams. He has done considerable consulting and research in the areas of modeling and applied statistics. He is currently a member of SIAM’s Education Committee and president of the Indiana Section of the Mathematical Association of America.